

Circular RNAs derived from *MIR156D* promote rice heading by repressing transcription elongation of pri-miR156d through R-loop formation

Received: 9 December 2023

Accepted: 27 February 2025

Published online: 25 March 2025



Yuan Su^{1,6}, Yong Yi^{1,6}, Shengchao Ge^{1,6}, Zi Wang¹, Zhangyu Wei¹,
Xueliang Liu¹, Chao Zhang², Qingjun Xie³, Haifeng Wang², Yangwen Qian⁴,
Bin Yu⁵✉ & Yunfeng Liu¹✉

In angiosperms, microRNA156 (miR156) acts as an intrinsic, endogenous developmental timer for the age-dependent transition from the juvenile to the adult phase^{1–3}. However, the mechanisms modulating the age-dependent expression pattern of miR156 are still poorly understood⁴. In this Article, we report that circular RNAs (ciMIR156Ds) derived from pri-miR156d negatively regulate miR156 levels in an aging-dependent manner in rice. The ciMIR156D levels increase as plants age, which is inversely correlated with the changes of pri-miR156d and miR156 abundance. Consistent with this observation, ciMIR156Ds deficiency caused by a spontaneous mutation increases pri-miR156d and miR156 levels, resulting in a delayed heading phenotype, whereas ciMIR156Ds overexpression has opposite effects, demonstrating that ciMIR156Ds are negative regulators of miR156. We further show that ciMIR156Ds form R-loops with *MIR156D* at the region where they derive in an aging-dependent manner, which reduces the occupancy of DNA-dependent RNA polymerase II at that location and hence impedes pri-miR156d elongation. These findings reveal a mechanism for regulating heading date by refining the aging-dependent expression of miR156.

Flowering timing or heading date stands as a pivotal agronomic trait in rice cultivation, representing a critical transition from vegetative to reproductive growth. It determines the production cycle, geographic and seasonal adaptability of rice^{5–7}. Agronomic yield traits such as spikelet number, grain yield per panicle and seed set are also associated with rice heading characteristics⁷. Therefore, the mechanisms regulating heading date hold notable importance in improving rice variety traits and enhancing yield potential. Aging is one of the crucial factors determining flowering time⁸, serving as an endogenous signal to suppress flowering during the juvenile phase while ensuring flowering during the adult phase without reliance on exogenous inductive factors⁹. The

microRNA156 (miR156) has been identified as a key signalling switch for the transition from vegetative to reproductive growth in the angiosperms and serves as a molecular marker of aging^{1–3}. By targeting members of the SQUAMOSA PROMOTER BINDING PROTEIN (SBP/SPL) family transcription factors for degradation, miR156 intricately regulates the expression of genes required for acquiring adult features, including the pivotal flower-timing gene *MIR172*, a target of SPLs. At the early developmental stage, high levels of miR156 repress SPL levels, maintaining the plant in the vegetative state. The miR156 levels decrease during shoot development, which leads to increased levels of SPLs, resulting in vegetative phase change. Although studies have

A full list of affiliations appears at the end of the paper. ✉e-mail: byu3@unl.edu; yunfengliu_bio@126.com

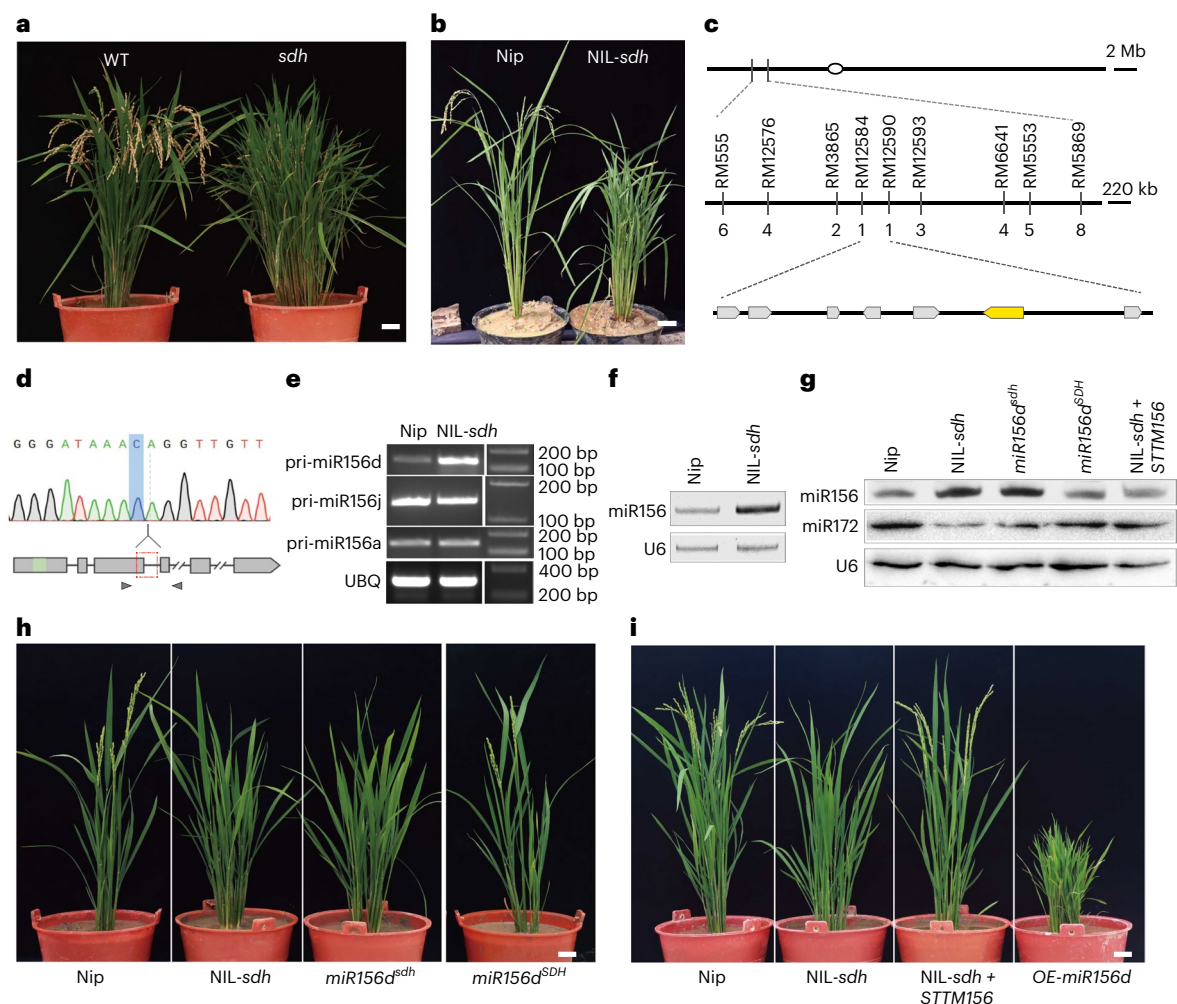


Fig. 1 | A deletion in *MIR156D* gene body increases the expression of miR156d and causes delayed heading date in *sdh*. **a,b**, Phenotypes of 98-day-old plants of wild-type plants (Gui649) and *sdh* mutants in the Gui649 genetic background (*sdh*) (**a**) and Nip and the near-isogenic line carrying *sdh* in the Nip genetic background (NIL-*sdh*) (**b**) grown under LD conditions. **c**, Diagram showing the *SDH* locus at the second chromosome. The numbers indicate the recombinants between the molecular markers. The grey boxes represent the genes located in the mapped region, while the yellow one indicates the coding region of pre-miR156d. **d**, The schematic diagram of a 133-bp nucleotide deletion within *MIR156D* in *sdh* mutant. Top: the DNA sequencing result of *sdh*. Bottom: the gene structure is represented by exons (boxes) and introns (lines). The green box in the first exon indicates the pre-miR156d coding region. The red dashed box indicates

the position of deleted nucleotides and the grey arrows represent the primers used for detecting *sdh* mutants. **e**, Transcript levels of pri-miR156s in Nip and NIL-*sdh* were analysed by RT-PCR. Ubiquitin (UBQ) was used as the control. **f,g**, The mature miR156 (**f**) or miR172 (**g**) were detected in the fifth leaves of various genotypic rice plants using RNA blotting. U6 was used as the loading control. **h,i**, Phenotypes of 91-day-old plants grown under LD conditions: Nip harbouring the *p156D:miR156^{sdh}* (*miR156^{sdh}*) and Nip harbouring the *p156D:miR156^{SDH}* (*miR156^{SDH}*) (**h**); and NIL-*sdh* harbouring the *p156D:STTM156* transgene (NIL-*sdh* + *STTM156*) and Nip harbouring the *p35S:MIR156D* transgene (*OE-miR156d*) (**i**). Phenotypic images were analysed by ImageJ v1.53t. Scale bars, 2 cm.

suggested that the aging-dependent decrease of miR156 abundance happens at transcriptional levels^{10–13}, the underlying mechanisms are largely unknown⁴.

Circular RNAs (circRNAs) are a class of non-coding RNAs containing the closed-loop structure. In animals, circRNAs play critical roles in regulating diverse biological processes via various functional mechanisms by interacting with proteins, RNAs and DNAs¹⁴. CircRNAs are also widely present in plants^{15,16}. However, their functions remain to be investigated.

From a population of Gui649, a *japonica* background restorer line (wild type, WT), we isolated the spontaneous mutant *semidwarf* and *delayed heading date* (*sdh*). Compared with WT, *sdh* exhibited semidwarf stature and delayed heading time; 10.6 days later under long-day (LD) conditions in Nanning, China (22° 50' N, 108° 17' E) during the summer season and 13.1 days later under short-day (SD) conditions in Lingshui, China (18° 31' N, 110° 03' E) during the winter season (Fig. 1a

and Extended Data Fig. 1a). Additionally, *sdh* had an increased number of tillers compared with WT (Extended Data Fig. 1b). The *sdh* appeared to be a single dominant mutation, given the observations that the F₁ plant resulting from the backcross between *sdh* and WT exhibited similar phenotypes to *sdh*, and that the ratio of plants displaying *sdh* phenotype to WT-like plants was ~2.5:1 (137:54, $\chi^2 = 1.09$, $P = 0.5–0.1$).

To identify the *sdh* mutation, we generated a near-isogenic line (NIL) of *sdh* (NIL-*sdh*) in the background of *japonica* rice Nipponbare (Nip) by backcrossing *sdh* to Nip seven times. NIL-*sdh* displayed similar phenotypes to those of the original *sdh* (Fig. 1b and Extended Data Fig. 1c–e) and was used for subsequent analyses. Bulk segregant analysis (BSA) of F₂ segregation plants displaying wild-type or *sdh* phenotype from the *sdh* × Nip cross showed that *sdh* was on the short arm of chromosome 2 (Extended Data Fig. 2a). Further map-based cloning analysis using a population derived from crossing NIL-*sdh* with the *indica* variety 9311 narrowed *sdh* to an 85.8-kilobase (kb) region (Fig. 1c),

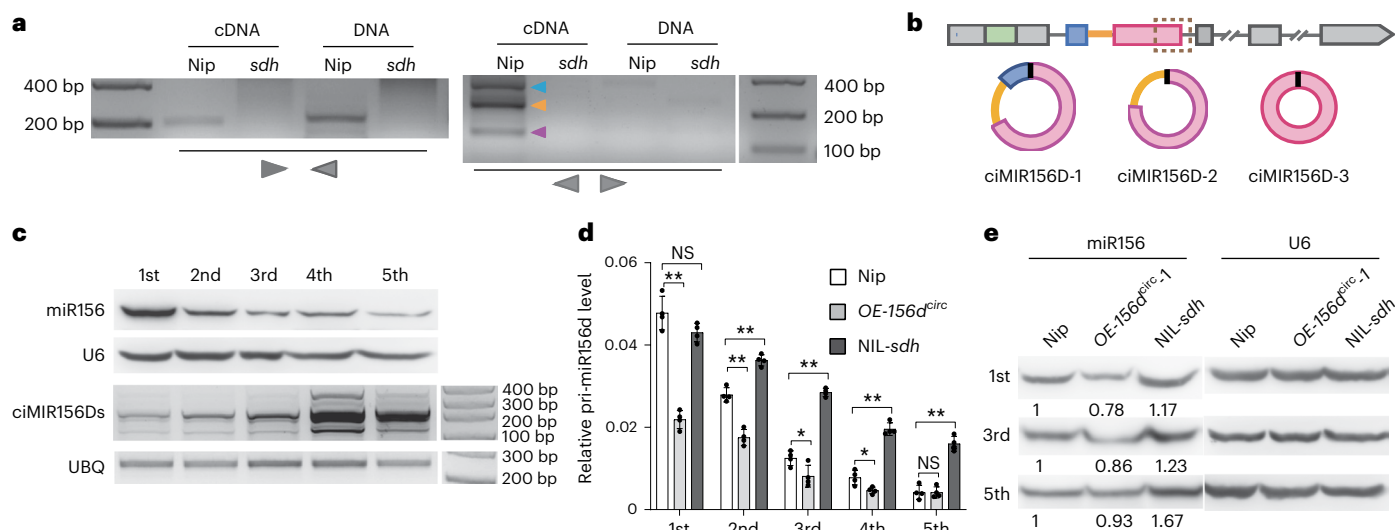


Fig. 2 | CircRNAs derived from the *sdh* deletion region regulate the transcription of miR156. **a**, Agarose gel electrophoreses showing circRNA amplicons from RNase R-treated RNA extracted from Nip and NIL-*sdh* plants.

Left: convergent primers were designed within the circRNA region and applied on both cDNA and genomic DNA samples of Nip and NIL-*sdh* plants. Right: divergent primers were used to amplify ciMIR156D-1, -2, -3 from cDNA and genomic DNA of Nip. The cDNA was produced from total RNA treated with RNase R and subject to reverse transcription. The blue, red and purple arrows indicate ciMIR156D-1, ciMIR156D-2 and ciMIR156D-3, respectively. Grey arrows indicate the convergent primers and divergent primers. **b**, Schematic diagram of three circRNAs generated from the second exon to the third exon of *MIR156D*. The gene structure is represented by exons (boxes) and introns (lines) and the green box in the first exon indicates the pre-miR156d coding region. The blue and pink boxes and yellow lines represent exons and introns, respectively, encompassing

the circRNAs. Black lines indicate the BSJ of circRNAs. The dashed box indicates the deleted DNA region in the *sdh* mutant. **c**, RNA blot analysis of miR156 levels from the first to the fifth leaf blades. U6 was used as a loading control (the top two lines). The expression of ciMIR156Ds from leaves at various stages was detected by RT-qPCR after RNase R treatments and gel electrophoreses. UBQ was used as a normalization control (the bottom two lines). The ciMIR156Ds, circRNAs derived from *MIR156D*. **d**, The transcript levels of pri-miR156d in leaves of various genotypes determined by RT-qPCR. The relative transcript abundance is compared to the level of ubiquitin. Data are shown as mean $\pm$ s.d. of four technical replicates. * $P < 0.05$; ** $P < 0.01$; NS, not significant using single-sided Student's *t*-tests. **e**, RNA blot analysis of miR156 levels in leaves of three stages from various genotypes. U6 was used as a loading control. Band intensity was analysed by Quantity One 1-D v.4.6.8 (Bio-Rad).

which contains six protein-coding genes and the non-coding *MIR156D* gene (*Os02g0180800*) (Extended Data Fig. 2b). Sequence analyses detected a 133-base pair (bp) nucleotide deletion (4512043–4512175) across the junction of the third exon and third intron of *MIR156D*, which was positioned at 710-bp downstream of the stem-loop coding region (Fig. 1d and Extended Data Fig. 2c). This deletion cosegregated with the *sdh* phenotypes and increased both pri-miR156d and miR156 levels in the flag leaf of NIL-*sdh* without altering the levels of other pri-miR156s relative to Nip (Fig. 1e,f and Extended Data Fig. 2d). Moreover, the transcript levels of *SPLs*, which are key regulators of flowering and tillering in rice^{17,18} and targets of miR156, were much lower in NIL-*sdh* compared with Nip (Extended Data Fig. 3a), suggesting that the deletion may cause developmental defects by increasing miR156 levels. To support this notion, we expressed a mutated copy of *MIR156D* harbouring the 133-bp deletion (*miR156^{sdh}*) and a WT copy of *MIR156D* (*miR156^{SDH}*) driven by its native promoter in Nip. Expression of *miR156^{sdh}*, but not *miR156^{SDH}*, increased accumulation of pri-miR156 and mature miR156 (Fig. 1g and Extended Data Fig. 3b), resulting in developmental defects similar to those of NIL-*sdh* (Fig. 1h and Extended Data Fig. 3c,d). In addition, we expressed an miR156 short tandem target mimic (STTM156) driven by the *MIR156D* promoter (Extended Data Fig. 3e) in NIL-*sdh*. Expression of STTM156 restored mature miR156 levels and rescued the developmental defects of NIL-*sdh*, when compared with Nip (Fig. 1i and Extended Data Fig. 3c,d,f). Furthermore, overexpression of *MIR156D* under the control of the 35S promoter dramatically delayed the heading date and increased tiller numbers in Nip (Fig. 1i and Extended Data Fig. 3c,d). Taken together, these results demonstrate that the 133-bp deletion increases miR156 levels, resulting in the developmental defects of *sdh*.

Next, we sought to elucidate how *sdh* alters *MIR156D* expression. We noticed that there are seven predicted circRNAs in the *MIR156D*

(ciMIR156Ds) coding region^{19–23} (Supplementary Data 1). Among them, we confirmed that three of these circRNAs—*Os_cir2088* (ciMIR156D-1), *OSA_circ02437* (ciMIR156D-2) and *OSA_circ02436* (ciMIR156D-3)—were expressed (Fig. 2a,b, Extended Data Fig. 4a and Supplementary Table 1). Because the *sdh* deletion spans the back-splicing junction (BSJ) that is essential for generating circRNAs from pri-miR156d (ref. 14), we suspected that it may impair circRNA production. Indeed, ciMIR156Ds were undetectable in *sdh* (Fig. 2a). This result led us to hypothesize that ciMIR156Ds may negatively regulate pri-miR156d expression. Indeed, the transcript levels of ciMIR156Ds were increased from the first to fifth leaves (peaked at the fourth leaf), which is reversibly correlated with changes of miR156 levels^{24,25} (Fig. 2c). Moreover, pri-miR156d levels were much higher from the second to fifth leaves in NIL-*sdh* than in Nip (Fig. 2d). Furthermore, expression of a DNA fragment covering the first exon to fourth exon (*OE-156d^{circ}*) of *MIR156D*, in which miR156 and miR156* sequences were mutated so that miR156 cannot be produced from this transgene, greatly increased ciMIR156D levels (Extended Data Fig. 4b,c). Increased ciMIR156D transcripts repressed the expression of pri-miR156d (Fig. 2d and Extended Data Fig. 5a) and miR156 (Fig. 2e), resulting in increased transcript levels of some *SPLs* genes (Extended Data Fig. 5c). It should be noted that despite the notably reduced pri-miR156d levels in the flag leaf (Extended Data Fig. 5a), miR156 levels in the flag leaf of *OE-156d^{circ}* plants did not decrease as much as pri-miR156d compared with Nip (Extended Data Fig. 5b), probably because miR156 levels may have reached the lowest threshold at this stage or due to the presence of other *MIR156* genes. In agreement with this, miR172 levels were increased at the flag leaf of the *OE-156d^{circ}* lines (Extended Data Fig. 5b). As a control, we overexpressed a truncated linear ciMIR156D (*OE-156d^{wt/3rd intron}*), in which the third intron was removed to block ciMIR156D production. *OE-156d^{wt/3rd intron}*, whose transcript levels were similar to those from *OE-156d^{circ}*, had no obvious

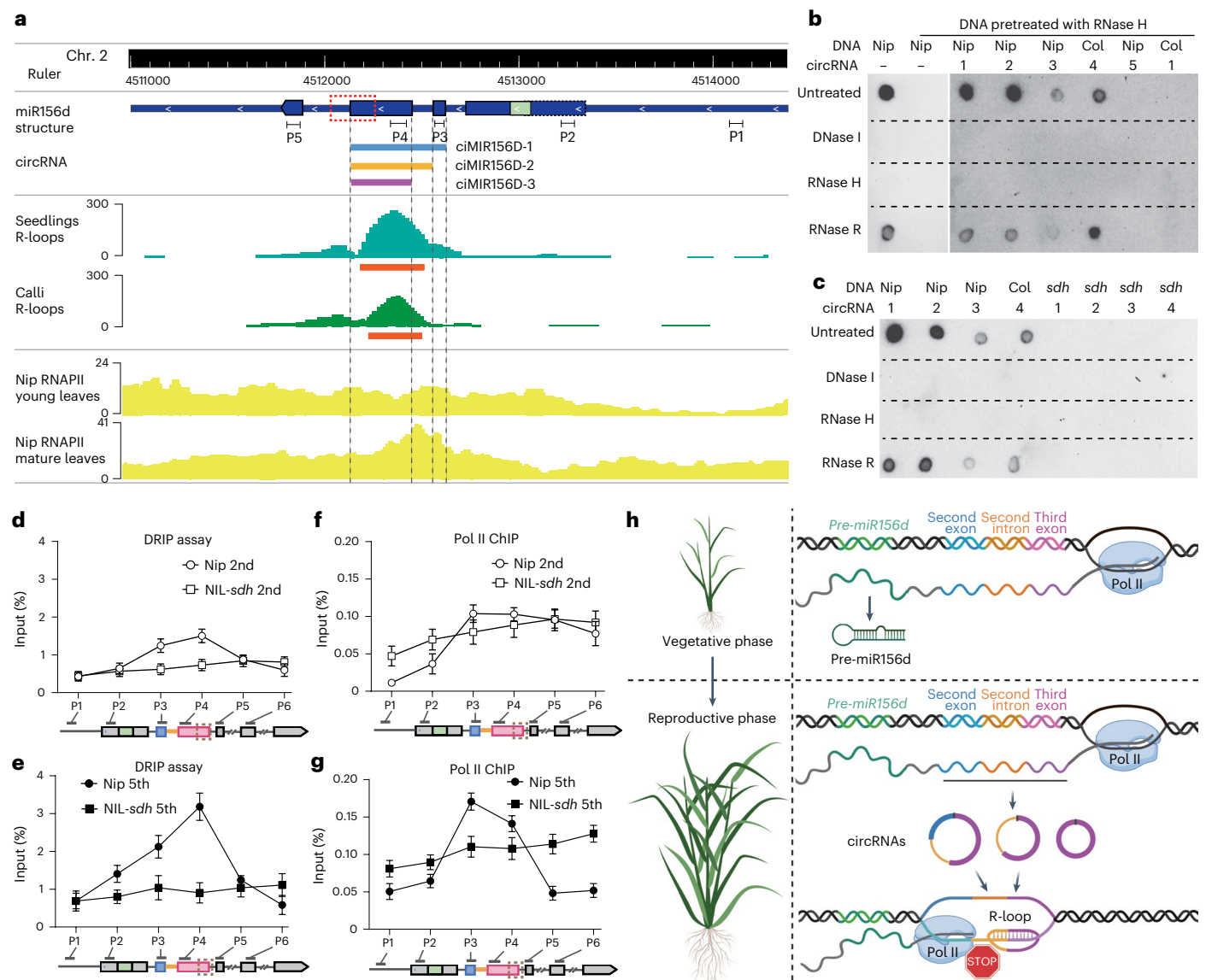


Fig. 3 | The formation of ciMIR156D-mediated R-loops hinders transcription elongation of *MIR156D*. **a**, Diagram of the localization of ciMIR156Ds, potential R-loops and the occupancy of Pol II at the *MIR156D* locus. The blue, yellow and purple lines indicate the positions of ciMIR156Ds at the *MIR156D* locus. The orange line indicates the peak of the R-loop at the *MIR156D* locus. R-loop peaks from seedlings (cyan) and calli (green) of Nip and the occupancy of Pol II at the *MIR156D* locus in young and mature leaves of Nip (yellow) were generated on the basis of the DRIP-seq and Pol II ChIP-seq data from refs. 31,34, respectively. The red dashed box indicates the nucleotide deletion in *sdh*. P1–P5 refer to the selected regions across the *MIR156D* gene for assays in d–g. **b, c**, Detection of circRNA–DNA hybrids by in vitro dot-blot assays with or without treatments of various nucleases in the set of WT Nip and Col (**b**) and the set of WT and *sdh* mutant (**c**). The purified circular circRNAs (1, ciMIR156D-1; 2, ciMIR156D-2; 3, ciMIR156D-3; 4, *AtSEPs* exon 5 circRNA; 5, circSg05160) were incubated with genomic DNA from Nip, *Arabidopsis* (Col) or NIL-*sdh* (*sdh*). The DNAs were pretreated with RNase H to remove pre-existing R-loops as indicated in (**b**) and all samples in (**c**). The R-loops were detected using S9.6 antibody recognizing the RNA–DNA hybrid.

effect on levels of pri-miR156d and miR156, as well as the heading date (Extended Data Figs. 5a,b and 6). Taken together, these results demonstrate that ciMIR156Ds represses the expression of *MIR156D*.

Beside circRNAs, back-splicing could also induce nonsense-mediated decay (NMD) which may affect pri-miR156d levels. To rule out this possibility, we generated a loss-of-function mutant of Os07g0495900, which encodes the orthologue gene of *Arabidopsis*

LBA1 (At5g47010) required for NMD^{26,27} and found that pri-miR156d and miR156 accumulation was not altered in this mutant relative to WT (Extended Data Fig. 7). Next, we asked how ciMIR156Ds repress miR156 levels. They could not function as an miRNA sponge because of the lack of a miR156 binding site or act on pri-miR156d RNA since they are derived from the same strand and do not have predicted complementary binding regions. Previous

reports have shown that certain nuclear-localized circRNAs can alter the expression of their parent genes^{28,29}. Because ciMIR156Ds are also localized in the cell nucleus (Extended Data Fig. 8a), we reasoned that they might affect pri-miR156d transcription. On the basis of the observation that some circRNAs bind their cognate DNA to form a RNA–DNA hybrid or R-loop to modulate gene expression³⁰, we next asked if ciMIR156Ds could form an R-loop with their cognate DNA. Indeed, the R-loops in the region spanning the second intron to third exon of *MIR156D* were found in data generated by rice DNA–RNA immunoprecipitation followed by sequencing (DRIP-seq)³¹ (Fig. 3a). Thus, we used dot blot with R-loop-specific antibodies to test if ciMIR156Ds could form R-loops in vitro. Incubation of R-loop free genomic DNA with ciMIR156D-1, ciMIR156D-2, ciMIR156D-3 and the positive control *AtSEP5* exon 5 circRNA³⁰, but not the negative control circRNA5g05160 (ref. 22), generated signals (Fig. 3b). These signals were R-loop specific because they became undetectable after treatments with DNase I, which degrades DNA, or RNase H, which degrades RNA in RNA–DNA hybrids. Moreover, treatment with RNase R, which removes linear RNAs, showed that 49.8%, 45.5% and 34.7% of the total R-loop signal were generated by ciMIR156D-1, ciMIR156D-2 and ciMIR156D-3, respectively (Fig. 3b), suggesting that ciMIR156D-1 and ciMIR156D-2 exhibited a strong ability to form R-loops with their cognate DNAs, while ciMIR156D-3, which contains only the third exon, showed a weaker binding capacity. Notably, incubation of ciMIR156Ds with R-loop-free DNAs from *sdh*, which still contain substantial sequences complementary to ciMIR156Ds, did not generate R-loop signals (Fig. 3c), suggesting that the 133-bp deletion in *sdh* is crucial for R-loop formation. Supporting this result, ciMIR156D-3, which partially overlaps with the deleted region, formed a weaker R-loop with wild-type genomic DNA (Fig. 3b,c). Taken together, these results demonstrate that ciMIR156Ds form R-loops with the DNA regions where they derive.

Next, we used DRIP with antibodies recognizing RNA–DNA hybrids followed by quantitative polymerase chain reaction (qPCR) to validate in vivo ciMIR156D-mediated R-loop formation in the second and fifth leaves. DRIP detected the enrichment of a circRNA–DNA R-loop in the region spanning both second intron and third exon of *MIR156D* in Nip but not in *sdh* (Fig. 3d,e). Moreover, the R-loop signals were higher in the fifth leaf than in the second leaf, which is consistent with the elevated ciMIR156D levels in respective leaves (Fig. 3d,e). Together with the in vitro results, these data reveal that ciMIR156D-1 and -2 form an R-loop with their cognate DNA in vivo.

Next, we asked how the R-loop formed by ciMIR156Ds modulates pri-miR156 transcription. In plants, R-loops formed by circRNA–DNA pairing have been shown to modulate epigenetic modifications^{31,32}. However, *sdh* did not substantially alter the levels of histone methylation or acetylation in either the miR156d coding region or the proximal region (Extended Data Fig. 8b,c). R-loops formed in gene bodies also can restrain RNA polymerase II (Pol II) extension rate during the elongation³³. Indeed, chromatin immunoprecipitation (ChIP) data of Pol II showed that Pol II occupancy at the R-loop region of mature leaves (flag leaves)³⁴, which had high ciMIR156D levels, was higher than that in ciMIR156D-low young leaves (seedling leaves) (Fig. 3a), suggesting that ciMIR156Ds may affect pri-miR156d elongation. To support this, we compared the Pol II occupancy at the *MIR156D* gene region in Nip and NIL-*sdh* in leaves of different stages. In the R-loop-rich fifth leaf (Fig. 3e), Pol II showed notably higher occupancy at the R-loop region in NIL-*sdh* relative to NIP (Fig. 3g). In contrast, in the R-loop-limited second leaf (Fig. 3d), the difference of Pol II occupancy between Nip and NIL-*sdh* was not substantial (Fig. 3f). These results reveal that ciMIR156D-1 and -2 affect pri-miR156d elongation in an R-loop-dependent manner.

As the major contributor for miR156 production (Extended Data Fig. 9) in rice, the fluctuation of miR156d transcripts shall substantially impact on the miR156 level. In this study, we identified ciMIR156Ds derived from pri-miR156d as key regulators of heading date in rice. Two out of three ciMIR156Ds bind their cognate DNA locus, forming R-loops. The R-loops slow the transcriptional elongation of pri-miR156d,

ultimately resulting in reduced accumulation of miR156. Notably, ciMIR156D levels increase from the first to fifth leaves, enabling them to modulate miR156 levels in an aging-dependent way. Indeed, lack of ciMIR156Ds causes delayed head date while overexpression of ciMIR156D results in earlier flowering time. How the ciMIR156D levels are modulated in an aging-dependent manner remains to be investigated. Intriguingly, circRNAs derived from MIR156 locus are also present in *Arabidopsis* and overlap with the R-loop at the region (Extended Data Fig. 10 and Supplementary Data 2), indicating that circRNA-mediated modulation of miR156 maybe a conserved mechanism to control the transition from the vegetative to reproductive stages. It has been reported that R-loops, formed near the transcription start site region of *MIRNAs*, promote cotranscriptional pri-miRNA processing³⁵. This raised a possibility that R-loops formed by ciMIR156s may affect pri-miR156d processing, in addition to their role in transcription regulation. However, these R-loops were formed after pre-miR156d. Thus, their effect on pri-miR156d processing remains to be investigated.

Multiple circRNAs have been shown to function through R-loop formation^{30,36}, suggesting that they may extensively affect gene transcription through mediating R-loop formation. In agreement with this, we identified 460 circRNAs originating from 175 miRNA loci from rice circRNA database²³ (Supplementary Data 3). Among them, 165 circRNAs from 78 miRNA loci potentially overlap with R-loops (Supplementary Data 4), suggesting that circRNA-mediated R-loop formation maybe a common mechanism in regulating miRNA levels.

Methods

Plant materials

To develop the NILs of *sdh*, *sdh* was consecutively backcrossed (BC) seven times with Nipponbare. The plants resulting from self-pollination of the BC7 plants carrying *SDH/sdh* were used as NILs. The plant population resulting from self-pollination of BC3 plants carrying *SDH/sdh* (BC3F2) was used for BSA assay.

Mapping of the *sdh* mutation

NIL-*sdh* was backcrossed with *indica* variety 9311 three times. The resulting BC3 plant carrying *SDH/sdh* (BC3F2) was self-pollinated to generate the fine mapping population. Using >3,000 BC3F2 plants, the *SDH* locus was mapped to a region between markers RM555 and RM5869 on chromosome 2, and further narrowed to an ~85.8-kb region flanked by markers RM12584 and RM1259. All plants were grown in the experimental field in Guangxi University in Nanning of Guangxi province or Lingshui of Hainan province, China. The sequences of the predicted genes in the candidate region were compared among the mapping parents. Markers used for genotyping are listed in Supplementary Data 5.

Transgene constructs

The 2,423-bp promoter of *MIR156D* was cloned into *pRHE* vector by in-fusion cloning to generate the *pRHE-pro156d* construct. The genomic region of *MIR156D* from first exon to fifth exon was amplified from DNA of Nip or NIL-*sdh* and fused with the sixth exon using fusion PCR to obtain the *miR156d^{SDH}* or *miR156^{sdh}* fragments, respectively. The *miR156d^{SDH}* and *miR156^{sdh}* were cloned into *pRHE-pro156d*, to generate *pRHE-pro156d::miR156^{SDH}* and *pRHE-pro156d::miR156^{sdh}*, respectively. The STTM targeting miR156 was amplified using PCR and cloned into *pRHE-pro156d*. The genomic region of *MIR156D* from first exon to fifth exon was mutated by site-directed mutagenesis and was cloned into pRHV under the control of the ubiquitin promoter to generate ciMIR156D overexpression construct.

RNA blots

RNA blot was performed as described in ref. 25. Briefly, after treatment with DNase I, 30 µg of total RNAs from various rice materials were separated by electrophoresis on 20% denaturing polyacrylamide gels,

transferred to a Hybond-N+ nylon membrane (Amersham Bioscience, GE Healthcare) and ultraviolet-crosslinked. After hybridization at 42 °C in ULTRAhyb-oligo hybridization buffer (Ambion) with a biotin-labelled (Invitrogen) probe, the membranes were washed twice at 40 °C in 2× SSC buffer and 0.5% SDS buffer for 30 min and for further biotin detection. Total RNA was extracted using the TRIzol reagent (Invitrogen) from sequentially developed leaves (first to fifth leaves) or flag leaf of 5-week-old rice.

In vitro RNA transcription, circularization and purification

In vitro RNA transcription, circularization and purification were performed as described in ref. 37 with slight modifications. A total of 1 µg of PCR-amplified DNA fragments with T7 promoter were incubated with 2 µl of T7 RNA polymerase enzyme (Promega), 0.5 mM dNTPs, RNase inhibitor and 2 mM GMP to produce linear RNAs. After DNase I treatment, 50 µg of in vitro transcribed RNAs were incubated with T4 RNA ligase 2 (NEB) for in vitro circularization. Following elimination of linear RNAs with RNase R digestion, the resulting circRNAs were purified with phenol–chloroform extraction and precipitated with ethanol. The purified circRNA was validated by PCR with divergent primers listed in Supplementary Data 5.

CircRNA amplification, Sanger sequencing and quantitative analysis of circRNA

The divergent and convergent primers were designed for corresponding circRNA sequences and BSJ sites. After reverse transcription using random primers, controls were amplified using complementary templates generated from linear RNA with convergent primers and divergent primers were used to detect the circRNAs. PCR products were dissected from gel and purified using the gel extraction kit (Tiangen DP209) for sequencing. The accumulations of the ciMIR156Ds in total RNA of the fourth leaf of Nip were measured by droplet digital PCR using automatic digital PCR system DQ24 (Sniper).

DRIP assay

DRIP assay was carried out as described in ref. 32 with slight modifications. The nuclei isolated from rice leaves were used to extract genomic DNA through three rounds of phenol, phenol–chloroform and chloroform extraction followed by cold ethanol precipitation. Purified genomic DNA was sonicated into fragments of 100–300 bp. The 3-µg fragmented DNA was kept as inputs and 5-µg fragmented DNA with or without RNase H treatment was purified and incubated overnight at 4 °C with S9.6 antibody (1:200, Abcam ab234957) in IP buffer (10 mM NaPO₄, 140 mM NaCl, 0.05% Triton X-100) followed by incubation with 50 µl of protein A/G dynabeads). After washing with IP buffer three times, the beads were eluted with 300 µl of elution buffer (50 mM Tris-HCl pH 8.0, 10 mM EDTA buffer, 0.5% SDS) twice at 65 °C for 30 min. The S9.6 antibody-bound DNA was then purified using phenol–chloroform extraction and pelleted with ethanol. DNA was resuspended for PCR assay. The percentage of immunoprecipitated DNA was calculated as fold-change over the input.

Histone methylation and acetylation assay

The histone methylation and acetylation assays were carried out as described in ref. 38 with slight modifications. Briefly, 1 g of the rice material was crosslinked in 1% formaldehyde. Then, nuclei were isolated and resuspended in nuclear lysis buffer. Following this step, DNA was interrupted with ultrasonic crusher and diluted with fivefold ChIP dilution buffer. After prehybridization the protein A/G dynabeads, antibodies recognizing H3K27ac (1:500, Abcam ab4729) or H3K27me3 (1:500, Abcam ab6002) were used for immunoprecipitation at 4 °C. After washing, DNAs were eluted, subjected to decrosslink and purified with phenol–chloroform extraction. The resulting DNAs were then precipitated with isoamyl alcohol and resuspended for PCR assay.

In vitro R-loop formation and detection assay

Equal amount of in vitro synthesized circRNAs were incubated with the DNA template respectively, heated for 5 min at 65 °C in hybridization buffer (10 mM pH 7.5 Tris-HCl, 0.1 mM EDTA, varying NaCl concentrations) and slowly cooled down to room temperature for R-loop formation in vitro. Then 2 µl of reaction solution was spotted onto Hybond-N+ nylon membrane. After air-drying, the membrane was preblocked in 5% milk overnight at 4 °C and then incubated with S9.6 monoclonal antibody (1:1,000, Abcam ab234957), followed by incubation with goat anti-mouse-HRP secondary antibody (1:10,000, Abcam ab205719) to detect DNA–RNA hybrids loop. DNase I, RNase H and RNase R (NEB) treatments were performed according to manufacturer's instruction using a standard hybridization buffer (10 mM Tris-HCl, pH 7.5, 50 mM NaCl, 0.1 mM EDTA)³⁰. DNase I digests double-stranded DNA; the endoribonuclease RNase H degrades RNA in the RNA–DNA hybrids; the exoribonuclease RNase R degrades linear RNA, but not circular RNA.

Bioinformatics validation of circRNAs-R-loop

The circRNA data of rice downloaded from Plantcircbase (v.7.0, <http://ibi.zju.edu.cn/plantcircbase/index.php>)²³ and BEDTools (v.2.28.0, <https://bedtools.readthedocs.io/en/latest/index.html>)³⁹ were used to identify circRNAs located within a 2-kb range upstream and downstream of miRNAs. A total of 460 circRNAs were identified in this process. Subsequently, based on previously published rice R-loops peak data³¹, BEDTools were used to identify circRNAs that may potentially form R-loops with at least a 1-bp overlap between circRNA and R-loop peak, resulting in the identification of 165 circRNAs originating from 78 miRNAs, which may mediate R-loop formation.

Statistics and reproducibility

GraphPad Prism v.9.0 was applied for the statistical analysis. At least three independent repeats had been performed with similar results for semiquantitative PCR with reverse transcription, agarose gel electrophoreses and RNA blots. For phenotypic studies of transgenic lines, at least three independent lines were analysed with similar phenotypes.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

Publicly available datasets used in this study include PlantcircBase v.7.0 (<http://ibi.zju.edu.cn/plantcircbase/index.php>) and Bedtools v.2.28.0 (<https://bedtools.readthedocs.io/en/latest/index.html>). DRIP-seq data used for R-loop identification in this study are available via ref. 31 (accession no. GSE111944). Pol II ChIP-seq data used in this study are available via ref. 34. RNA-seq datasets of NIL-*sdh* and Nip are available via the Sequence Read Archive of the National Center for Biotechnology Information (accession nos. PRJNA1171578 and PRJNA1171724). All other data supporting the findings of this study are available in the paper and its Supplementary Information and Supplementary Data 1–6. Source data are provided with this paper.

References

- Wu, G. et al. The sequential action of miR156 and miR172 regulates developmental timing in *Arabidopsis*. *Cell* **138**, 750–759 (2009).
- Wang, J. W., Czech, B. & Weigel, D. miR156-regulated SPL transcription factors define an endogenous flowering pathway in *Arabidopsis thaliana*. *Cell* **138**, 738–749 (2009).
- Wang, J. W. Regulation of flowering time by the miR156-mediated age pathway. *J. Exp. Bot.* **65**, 4723–4730 (2014).
- Fouracre, J. P. & Poethig, R. S. The role of small RNAs in vegetative shoot development. *Curr. Opin. Plant Biol.* **29**, 64–72 (2016).
- Zeng, Q., Zhao, Z. H. & Zhao, S. Q. Signal pathways of flowering time regulation in plant. *Yi Chuan* **28**, 1031–1036 (2006).

6. Doi, K. et al. Ehd1, a B-type response regulator in rice, confers short-day promotion of flowering and controls FT-like gene expression independently of Hd1. *Genes Dev.* **18**, 926–936 (2004).
7. Xue, W. et al. Natural variation in Ghd7 is an important regulator of heading date and yield potential in rice. *Nat. Genet.* **40**, 761–767 (2008).
8. Freytes, S. N., Canelo, M. & Cerdán, P. D. Regulation of flowering time: when and where? *Curr. Opin. Plant Biol.* **63**, 102049 (2021).
9. Hyun, Y., Richter, R. & Coupland, G. Competence to flower: age-controlled sensitivity to environmental cues. *Plant Physiol.* **173**, 36–46 (2017).
10. Guo, C. et al. Repression of miR156 by miR159 regulates the timing of the juvenile-to-adult transition in *Arabidopsis*. *Plant Cell* **29**, 1293–1304 (2017).
11. Zhang, H. et al. Genome-wide mapping of the HY5-mediated gene networks in *Arabidopsis* that involve both transcriptional and post-transcriptional regulation. *Plant J.* **65**, 346–358 (2011).
12. Xu, L. et al. Regulation of rice tillering by RNA-directed DNA methylation at miniature inverted-repeat transposable elements. *Mol. Plant* **13**, 851–863 (2020).
13. Cui, Y. et al. The heterochronic gene *Oryza sativa* LIKE HETEROCHROMATIN PROTEIN 1 modulates miR156b/c/i/e levels. *J. Integr. Plant Biol.* **62**, 1839–1852 (2020).
14. Kristensen, L. S. et al. The biogenesis, biology and characterization of circular RNAs. *Nat. Rev. Genet.* **20**, 675–691 (2019).
15. Liu, R., Ma, Y., Guo, T. & Li, G. Identification, biogenesis, function, and mechanism of action of circular RNAs in plants. *Plant Commun.* **4**, 100430 (2023).
16. Chu, Q. et al. Characteristics of plant circular RNAs. *Brief. Bioinform.* **21**, 135–143 (2020).
17. Jiao, Y. et al. Regulation of OsSPL14 by OsmiR156 defines ideal plant architecture in rice. *Nat. Genet.* **42**, 541–544 (2010).
18. Miura, K. et al. OsSPL14 promotes panicle branching and higher grain productivity in rice. *Nat. Genet.* **42**, 545–549 (2010).
19. Lu, T. et al. Transcriptome-wide investigation of circular RNAs in rice. *RNA* **21**, 2076–2087 (2015).
20. Ye, C. Y., Chen, L., Liu, C., Zhu, Q. H. & Fan, L. Widespread noncoding circular RNAs in plants. *New Phytol.* **208**, 88–95 (2015).
21. Zhang, P. et al. PlantCircNet: a database for plant circRNA-miRNA-mRNA regulatory networks. *Database* **2017**, bax089 (2017).
22. Fan, J. et al. circRNAs are involved in the rice–*Magnaporthe oryzae* interaction. *Plant Physiol.* **182**, 272–286 (2020).
23. Xu, X. et al. PlantCircBase 7.0: full-length transcripts and conservation of plant circRNAs. *Plant Commun.* **3**, 100343 (2022).
24. Tanaka, N. et al. The COP1 ortholog PPS regulates the juvenile-adult and vegetative-reproductive phase changes in rice. *Plant Cell* **23**, 2143–2154 (2011).
25. Xie, K. et al. Gradual increase of miR156 regulates temporal expression changes of numerous genes during leaf development in rice. *Plant Physiol.* **158**, 1382–1394 (2012).
26. Yoine, M. et al. The *lba1* mutation of UPF1 RNA helicase involved in nonsense-mediated mRNA decay causes pleiotropic phenotypic changes and altered sugar signalling in *Arabidopsis*. *Plant J.* **47**, 49–62 (2006).
27. Kurihara, Y. et al. Genome-wide suppression of aberrant mRNA-like noncoding RNAs by NMD in *Arabidopsis*. *Proc. Natl Acad. Sci. USA* **106**, 2453–2458 (2009).
28. Chen, L. L. The biogenesis and emerging roles of circular RNAs. *Nat. Rev. Mol. Cell Biol.* **17**, 205–211 (2016).
29. Zhang, Y. et al. Circular intronic long noncoding RNAs. *Mol. Cell* **51**, 792–806 (2013).
30. Conn, V. M. et al. A circRNA from SEPALLATA3 regulates splicing of its cognate mRNA through R-loop formation. *Nat. Plants* **3**, 17053 (2017).
31. Fang, Y. et al. Characterization of functional relationships of R-loops with gene transcription and epigenetic modifications in rice. *Genome Res.* **29**, 1287–1297 (2019).
32. Xu, W. et al. The R-loop is a common chromatin feature of the *Arabidopsis* genome. *Nat. Plants* **3**, 704–714 (2017).
33. Shivji, M. K. K., Renaudin, X., Williams, C. H. & Venkitaraman, A. R. BRCA2 regulates transcription elongation by RNA polymerase II to prevent R-loop accumulation. *Cell Rep.* **22**, 1031–1039 (2018).
34. Xie, L. et al. RiceENCODE: a comprehensive epigenomic database as a rice encyclopedia of DNA elements. *Mol. Plant* **14**, 1604–1606 (2021).
35. Gonzalo, L. et al. R-loops at microRNA encoding loci promote co-transcriptional processing of pri-miRNAs in plants. *Nat. Plants* **8**, 402–418 (2022).
36. Liu, Y. et al. Back-spliced RNA from retrotransposon binds to centromere and regulates centromeric chromatin loops in maize. *PLoS Biol.* **18**, e3000582 (2020).
37. Li, X. et al. Linking circular intronic RNA degradation and function in transcription by RNase H1. *Sci. China Life Sci.* **64**, 1795–1809 (2021).
38. Huang, R. et al. BLISTER promotes seed maturation and fatty acid biosynthesis by interacting with WRINKLED1 to regulate chromatin dynamics in *Arabidopsis*. *Plant Cell* **34**, 2242–2265 (2022).
39. Quinlan, A. R. BEDTools: the Swiss-Army tool for genome feature analysis. *Curr. Protoc. Bioinformatics* <https://doi.org/10.1002/0471250953.bi1112s47> (2014).

Acknowledgements

We thank W. Zhang of Nanjing Agricultural University for providing R-loop results and Q. Song of Nanjing Agricultural University for analysing BSA results. This work was supported by BaGui Scholars Foundation of Guangxi (to Y.L.); National Natural Science Foundation of China (31970603 to Y.S.); National Institute of Health (GM127414 to B.Y.); and National Science Foundation (MCB-1818082 to B.Y.).

Author contributions

Y.S., Y.Y., S.G., Z. Wang, X.L. and Z. Wei performed the experiments and analysed the data. Y.S., S.G. and H.W. modelled and wrote the paper. C.Z., S.G. and Q.X. carried out data analysis. Y.Q. helped create rice mutants. Y.L. and B.Y. jointly supervised the research and wrote the paper. All authors provided feedback on the paper.

Competing interests

The authors declare no competing interests.

Additional information

Extended data is available for this paper at <https://doi.org/10.1038/s41477-025-01961-7>.

Supplementary information The online version contains supplementary material available at <https://doi.org/10.1038/s41477-025-01961-7>.

Correspondence and requests for materials should be addressed to Bin Yu or Yunfeng Liu.

Peer review information *Nature Plants* thanks the anonymous reviewers for their contribution to the peer review of this work.

Reprints and permissions information is available at www.nature.com/reprints.

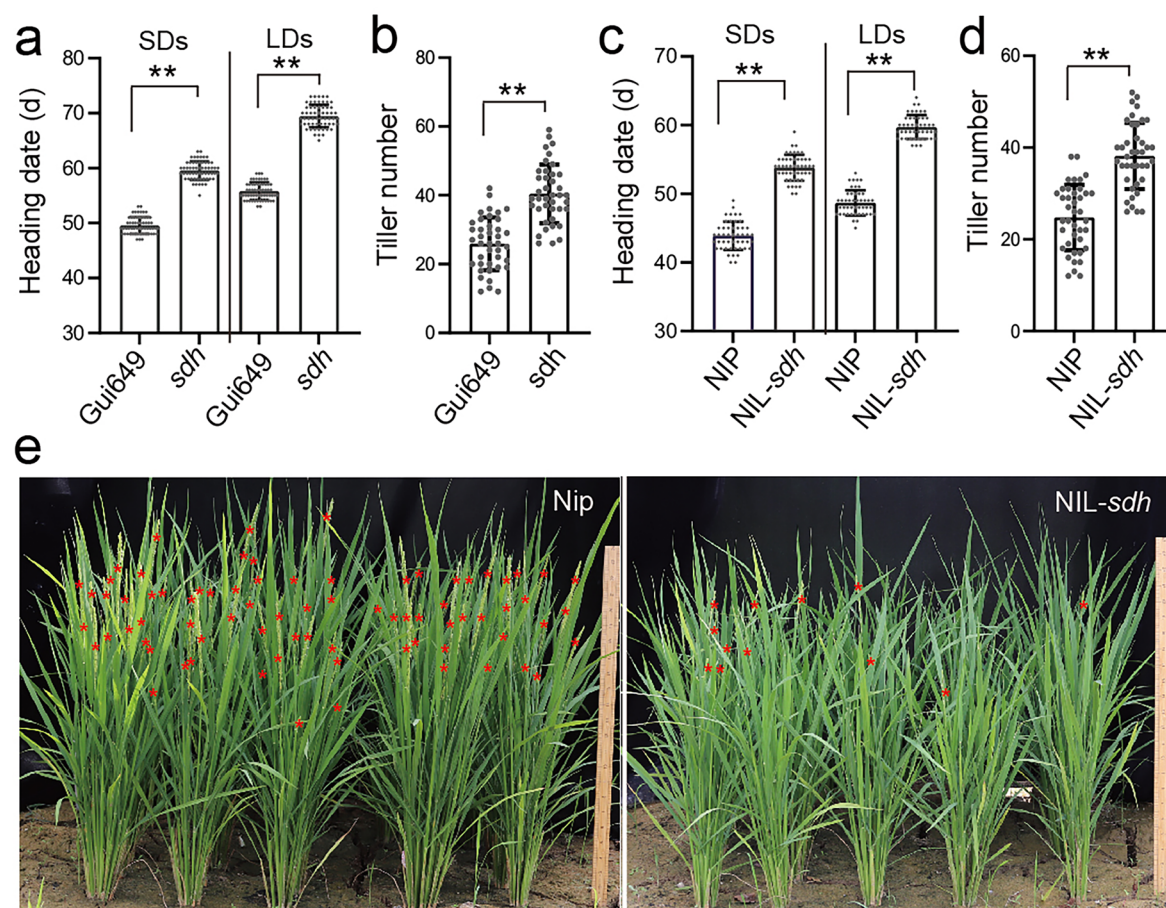
Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with

the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.

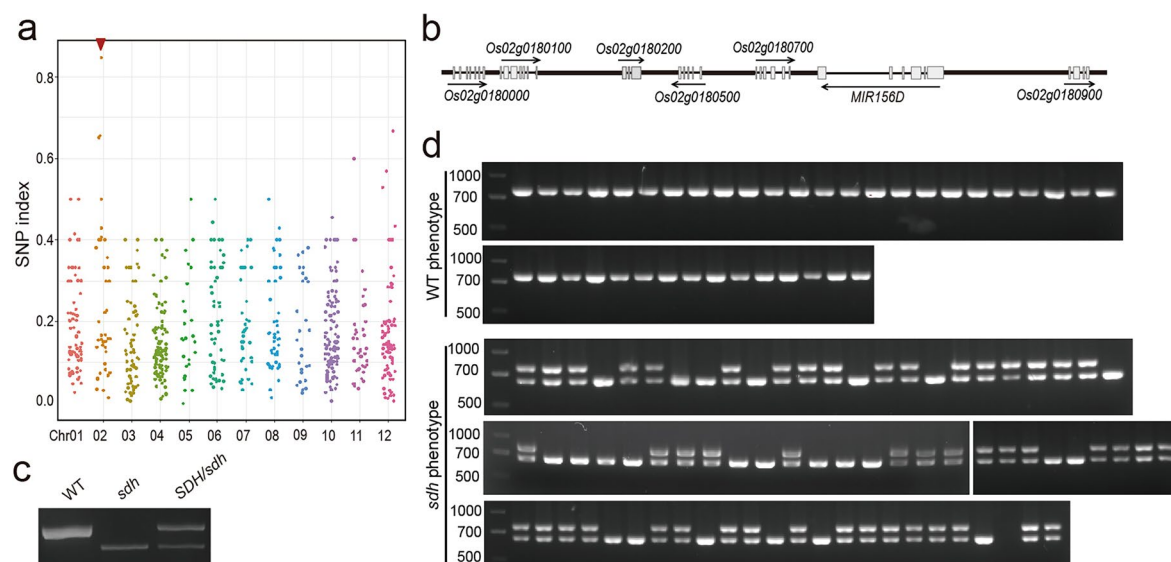
© The Author(s), under exclusive licence to Springer Nature Limited 2025

¹State Key Laboratory for Conservation and Utilization of Subtropical Agro-Bioresources, College of Life Science and Technology, Guangxi University, Nanning, China. ²State Key Laboratory for Conservation and Utilization of Subtropical Agro-Bioresources, College of Agriculture, Guangxi University, Nanning, China. ³State Key Laboratory for Conservation and Utilization of Subtropical Agro-Bioresources, Guangdong Provincial Key Laboratory of Plant Molecular Breeding, South China Agricultural University, Guangzhou, China. ⁴Wimi Biotechnology (Hainan) Co. Ltd, Sanya, China. ⁵Center for Plant Science Innovation and School of Biological Sciences, University of Nebraska, Lincoln, NE, USA. ⁶These authors contributed equally: Yuan Su, Yong Yi, Shengchao Ge. ✉e-mail: byu3@unl.edu; yunfengliu_bio@126.com



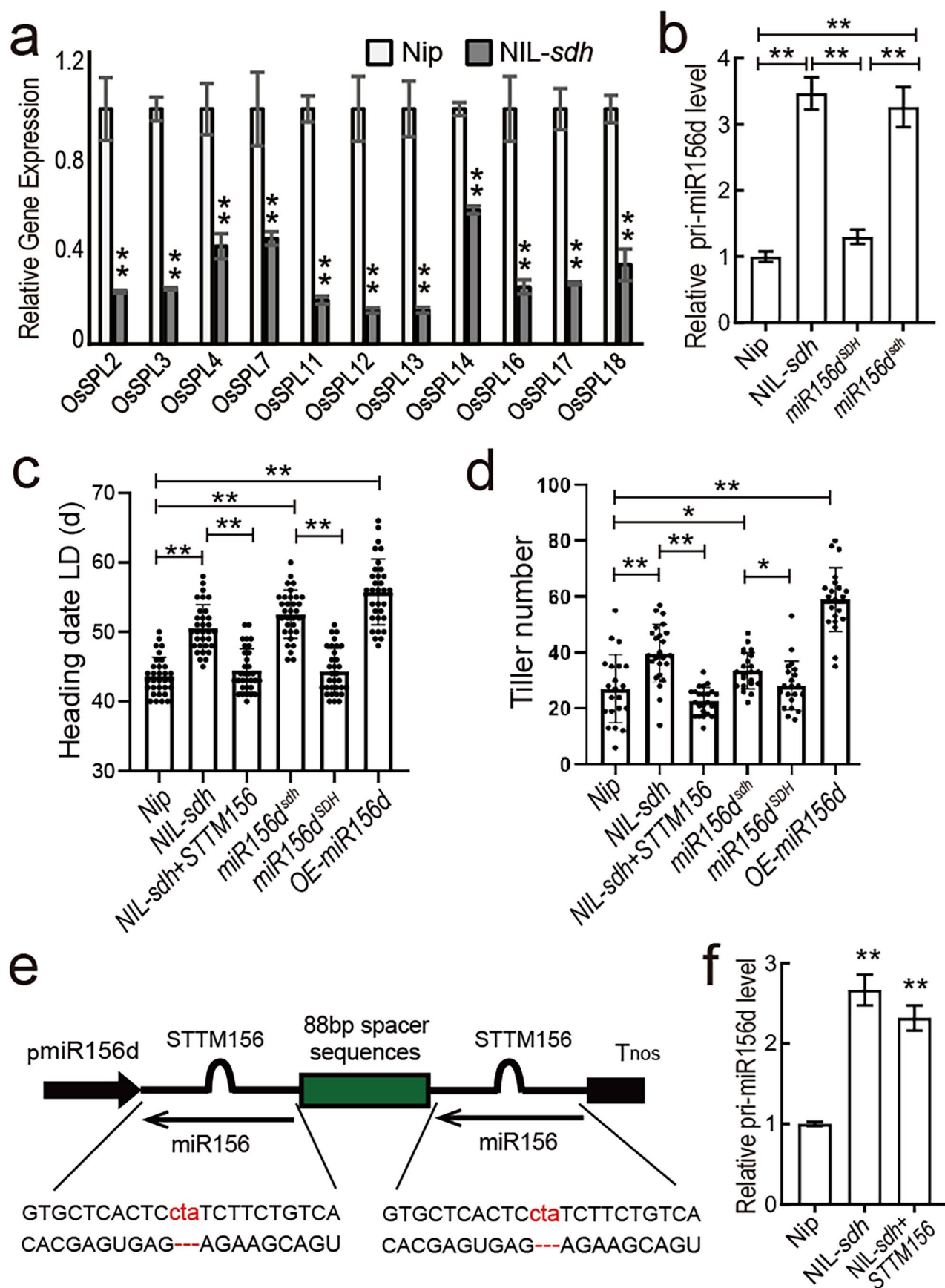
Extended Data Fig. 1 | The effect of the *sdh* allele on heading date. **a, c.** The heading date of indicated genotypes under long-days (LDs) and short-days (SDs). The days towards heading of each plant were recorded as the number of days beginning from transplanting seedlings to soil to the appearance of the first panicle. **b, d.** The tiller numbers of indicated genotypes under LDs. Values

in **a–d** are given as means $\pm$ SD. *, $P < 0.05$; **, $P < 0.01$ compared with the wild type by single-sided Student's *t*-test. **e.** Phenotypes of 98-day-old Nipponbare (Nip) and NIL-*sdh* plants grown under LD conditions in paddy fields of Nanning (China) during maturation period.



Extended Data Fig. 2 | Positioning cloning of *SDH*. **a**, Identification of genomic regions possibly harboring causative mutations using MutMap. The X-axis represents the chromosome positions. The Y-axis represents SNP index for the mutant. The red arrow indicates initial mutation locus interval. **b**, The diagram of predicted genes between markers RM12584 and RM12590. The boxes represent exons, while the slim lines and bold lines represent introns and regions between

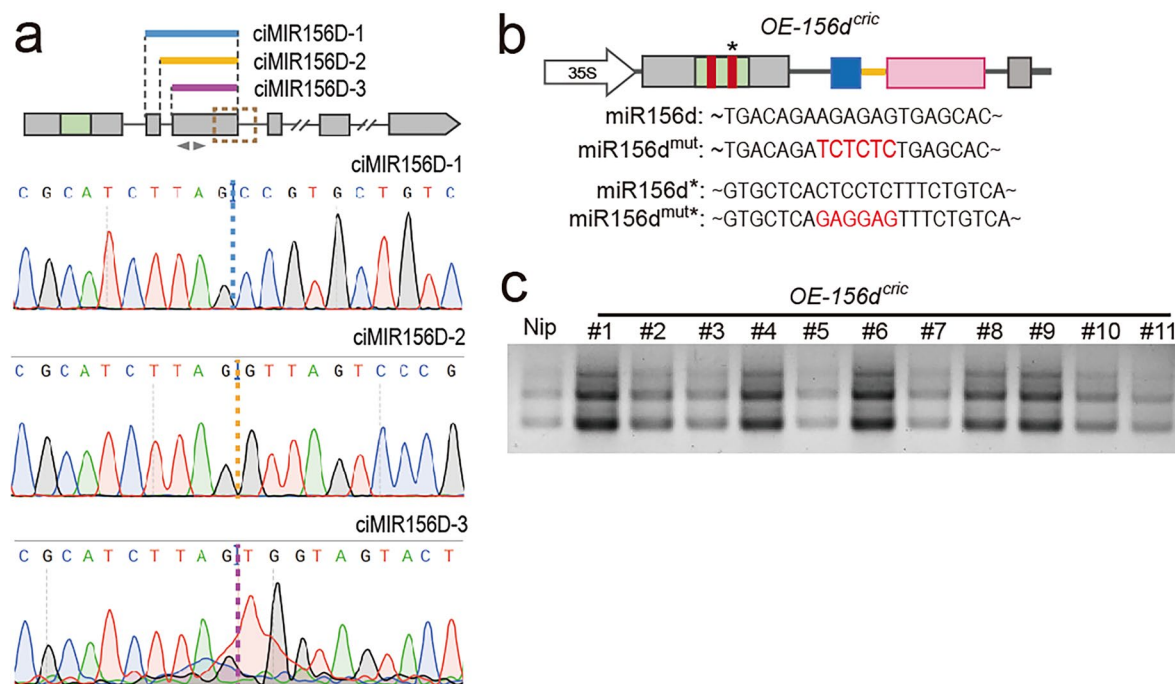
genes, respectively. **c**, PCR analysis of *sdh*. Primer positions were indicated in Fig. 1d. WT: Gui649; *sdh*: SDH/*sdh* obtained from Gui649 x *sdh* crossing. **d**, Association analysis between phenotype segregation and DNA deletion in *MIR156D* in F2 progenies generated from Gui649 and *sdh* crossing. The analyzed samples were collected based on WT or delayed heading phenotypes and subject to PCR reactions using primer shown in (Fig. 1d).



Extended Data Fig. 3 | See next page for caption.

Extended Data Fig. 3 | Functional verification of *SDH*. **a**, The relative expression levels of *SPL* genes in the flag leaves of Nip and NIL-*sdh* detected by RT-qPCR. **b**, The relative expression levels of pri-miR156d in the flag leaves of various genotypes detected by RT-qPCR. **c**, Heading dates of various genotypes under LDs. The days to heading of each plant were recorded as the number of days beginning from transplanting seedlings to soil to the appearance of the first panicle. **d**, The tiller numbers of various genotypic plants under LDs. Data in (**c**, **d**) are shown as means $\pm$ SD, *, $P < 0.05$ **, $P < 0.01$ by single-sided Student's

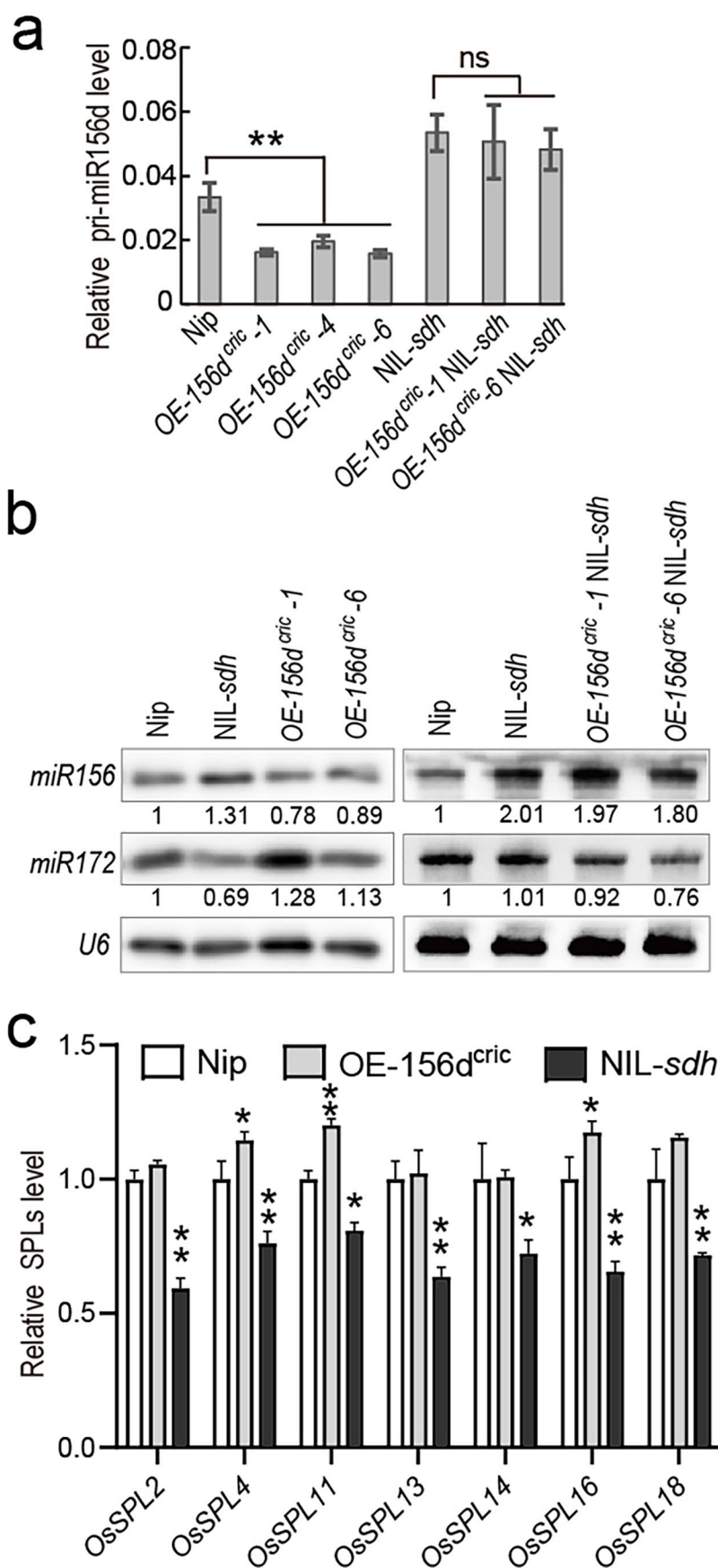
t-test. **e**, The diagram of structure of STTM, including the promoter of *MIR156D*, spacer linker between STTMs, and designed STTM156 in which a weak imperfect 'stem-loop' was included (marked by red). **f**, The relative expression levels of pri-miR156d in the flag leaves of various genotypes detected by RT-qPCR. Expression of Ubiquitin was used as an internal control, and the expression level of each gene in Nip was set as 1 in (**a**, **b**, **f**). Data are shown as means $\pm$ SD ($n = 3$), *, $P < 0.05$; **, $P < 0.01$ by single-sided Student's t-test.



Extended Data Fig. 4 | Identification and over-expression of ciMIR156Ds.

a, Diagram showing the positions of ciMIR156Ds at the *MIR156D* locus and their validation with divergent primers followed by Sanger sequencing. Dashed lines indicate in electropherogram represent reads matching the back-splicing junction. The typical locations of divergent primers used for circRNA-specific amplification from exon 3 by RT-PCR are shown as grey arrows. **b**, Schematic diagram of constructs used to over-express ciMIR156D (*OE-miR156d^{circ}*)

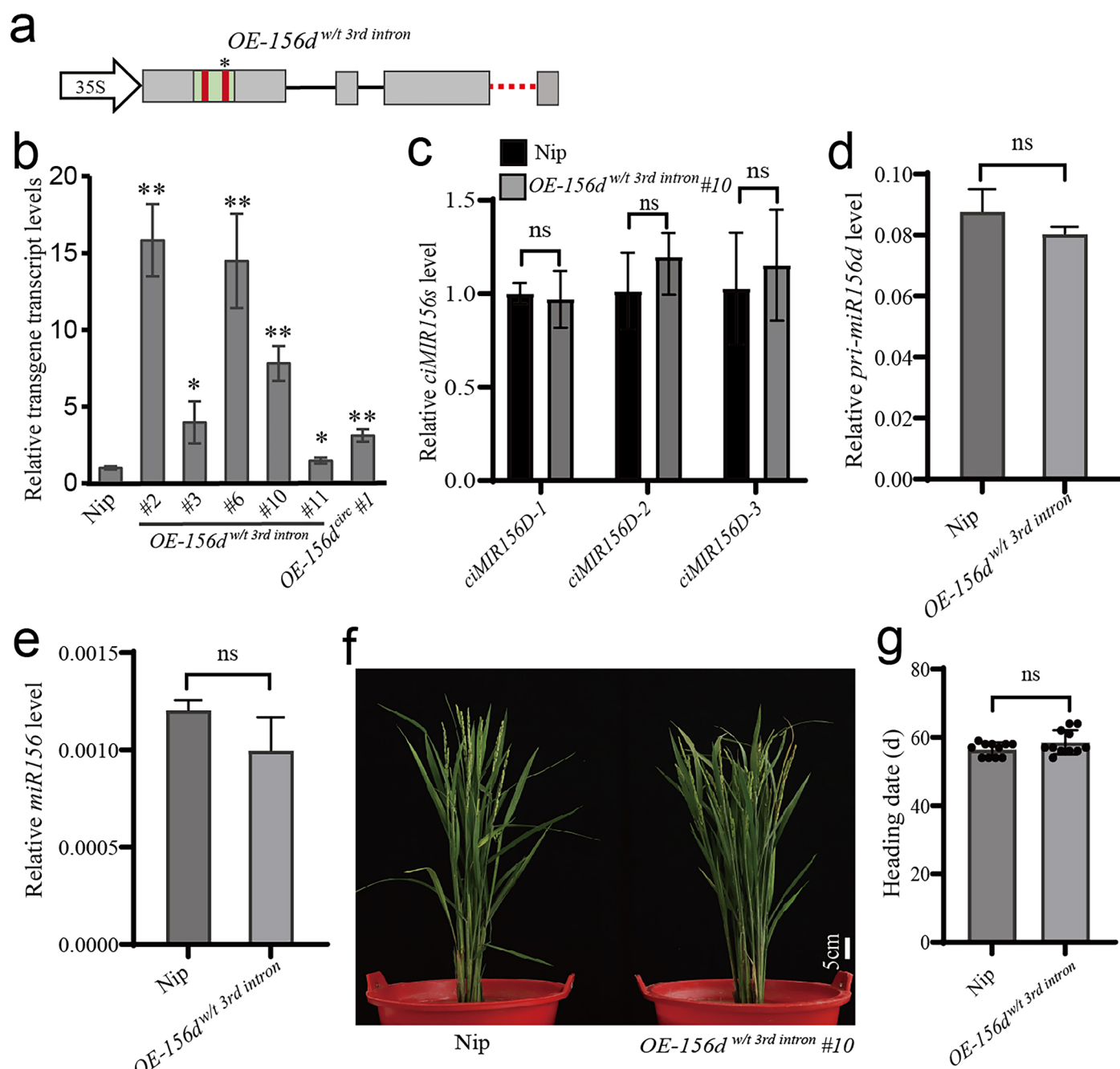
harboring a region from the 1st to the 4th exon of MIR156D. The red lines indicate that miR156 and miR156* were mutated in the construct. Mutated miR156d and miR156* sequences were shown below. **c**, ciMIR156D transcript levels in 11 independent *OE-miR156d^{circ}* transgenic plants detected by RT-PCR. Nip was used as a positive control. Line #1, #4 and #6 were selected for subsequent experiments as *OE-miR156d^{circ}* rice.



Extended Data Fig. 5 | See next page for caption.

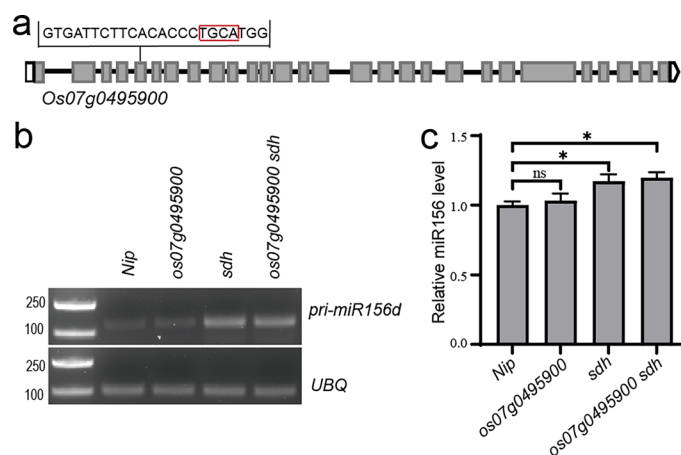
Extended Data Fig. 5 | The effects of circRNAs on expressions of miR156 and its target genes. **a**, The relative expression levels of pri-miR156d in the flag leaves of various genotypes detected by RT-qPCR using primers located at 5th exon. **b**, miR156 and miR172 were detected in flag leaves of various genotypes using Northern blot. U6 was used as the loading control. **c**, The relative expression

levels of *SPL* genes in the flag leaves of various genotypes detected by RT-qPCR. Ubiquitin was used as an internal control, and the expression levels of each gene in Nip were set as 1 in (**a**, **c**). Data are shown as means $\pm$ SD ($n = 3$), *, $P < 0.05$; **, $P < 0.01$; ns, not significant by single-sided Student's *t*-test.



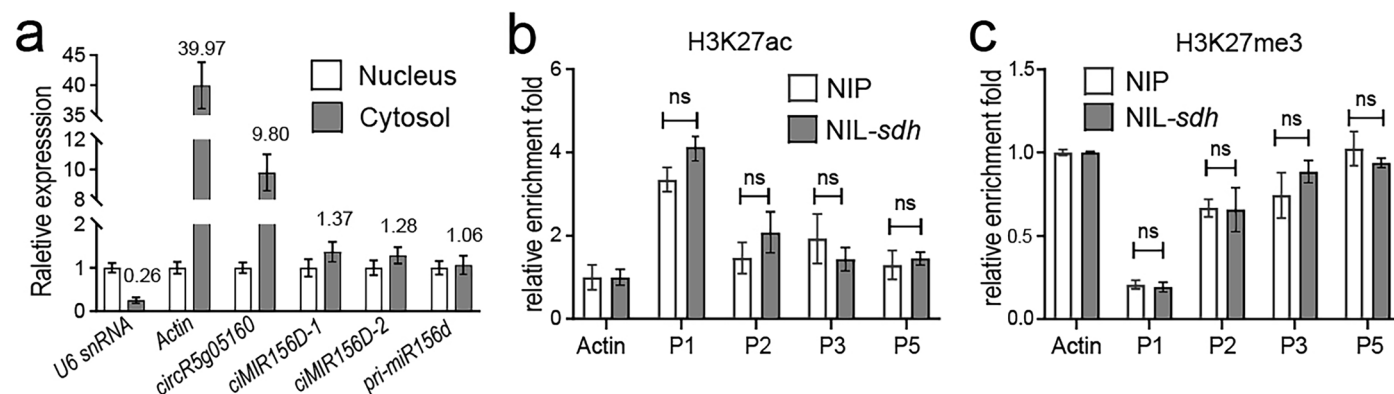
Extended Data Fig. 6 | The effects of linear RNAs of *OE-156d*^{w/t 3rd intron} on *miR156*. **a**, Schematic diagram of constructs used to over-express truncated *MIR156D* harboring a region from the 1st to the 4th exon of *MIR156D* without 3rd intron (*OE-156d*^{w/t 3rd intron}). The gene structure is represented by exons (boxes) and introns (lines), and the green box in the 1st exon indicates the pre-miR156 coding region. The red lines in 1st exon indicate that miR156 and miR156' were mutated in the construct, and mutated miR156d and miR156' sequences were shown in Supplementary Fig. 4b. The red dashed line indicates the deleted 3rd intron in the construct. **b**, The relative transgene transcript levels of *OE-156d*^{w/t 3rd intron} and *OE-156d*^{cim} in the flag leaves of various genotypes detected by RT-qPCR using

primers located at the 3rd exon. The transcript levels of *pri-miR156* in Nip were set as 1. The line #10 was used for the subsequent data. **c**, The relative expression levels of *ciMIR156D*s in the flag leaves of Nip and *OE-156d*^{w/t 3rd intron} #10 detected by RT-qPCR. **d**, The relative expression levels of *pri-miR156* in the flag leaves detected by RT-qPCR using primers located at 5th exon. **e**, The relative expression levels of *miR156* in the flag leaves among NIP and *OE-156d*^{w/t 3rd intron} #10 detected by stem loop RT-qPCR. **f, g**, The heading date of indicated genotypes under long days (LDs). *Ubiquitin* (b-d) or *U6* (e) was used as an internal control. Data are shown as means $\pm$ SD ($n = 3$), *, $P < 0.05$; **, $P < 0.01$; ns, not significant by single-sided Student's *t*-test.



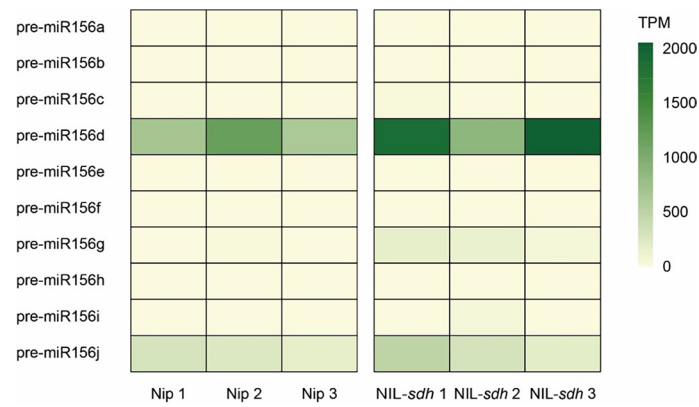
Extended Data Fig. 7 | The pri-miR156d of Nip and NIL-sdh were not affected by NMD. a, The representative diagram of *Os07g0495900* gene structure. Exons and introns are indicated by boxes and lines respectively. The *os07g0495900* mutant with a 4 bp deletion in the 5th exon was generated by CRISPR-Cas9 system. **b, c**, The relative levels of pri-miR156d (b) and miR156 (c) in Nip,

os07g0495900, NIL-*sdh* (*sdh*) and *os07g0495900 sdh* were detected by RT-PCR and stem-loop RT-qPCR, respectively. *Ubiquitin* (*UBQ*) and U6 were used as the control in b and c, respectively. Data are shown as means $\pm$ SD ($n = 3$), *, $P < 0.05$; ns, not significant by single-sided Student's *t*-test.

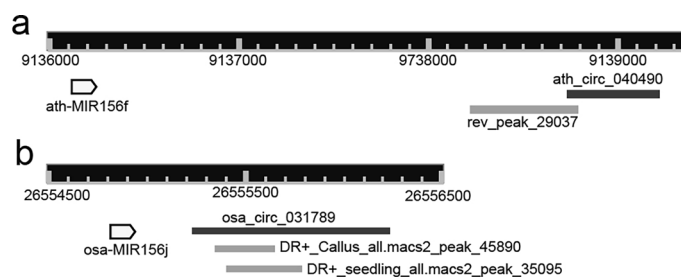


Extended Data Fig. 8 | The cellular localization of circRNAs and epigenetic modifications of *MIR156D* region. **a**, RT-qPCR analysis of circRNAs in the nuclear and cytosolic fractions in flag leaves of Nip. The nuclear abundance is set as 1. Data is shown as means $\pm$ SD (n = 3). **b,c**, The levels of H3K27ac (**b**) and H3K27me3 (**c**) at the *MIR156D* locus in wild-type and NIL-*sdh*. Four pairs of primers (used in Fig. 3a) around the deletion region were used to check the

enrichment of H3K27ac or H3K27me3 modifications. The flag leaves were collected for ChIP-qPCR assays. The levels of H3K27ac and H3K27me3 of Actin gene in wild-type was set to 1.0. Three biological replicates were performed. Data is shown as means $\pm$ SD (n = 3). A single-sided Student's t-test was used to generate the P values, *, P < 0.05; **, P < 0.01; ns, not significant.



Extended Data Fig. 9 | Transcript levels of individual miR156 members in Nip and NIL-sdh. The transcript levels of individual miR156 members in RNA-seq assay from 4th leaves of Nip and NIL-sdh were shown as TPM (Transcripts Per Kilobase of exon model per Million mapped reads) in heatmap, with three biological replicates for each sample.



Extended Data Fig. 10 | The circRNA-R-loop may exist in other *MIR156* loci of Arabidopsis and rice. a, b, Diagram showing the circRNAs derived from *MIR156F* locus of Arabidopsis (a) and *MIR156J* locus of rice (b) overlap with the R-loop.

Reporting Summary

Nature Portfolio wishes to improve the reproducibility of the work that we publish. This form provides structure for consistency and transparency in reporting. For further information on Nature Portfolio policies, see our [Editorial Policies](#) and the [Editorial Policy Checklist](#).

Statistics

For all statistical analyses, confirm that the following items are present in the figure legend, table legend, main text, or Methods section.

n/a	Confirmed
☐	☒ The exact sample size (<i>n</i>) for each experimental group/condition, given as a discrete number and unit of measurement
☐	☒ A statement on whether measurements were taken from distinct samples or whether the same sample was measured repeatedly
☐	☒ The statistical test(s) used AND whether they are one- or two-sided <i>Only common tests should be described solely by name; describe more complex techniques in the Methods section.</i>
☒	☐ A description of all covariates tested
☒	☐ A description of any assumptions or corrections, such as tests of normality and adjustment for multiple comparisons
☐	☒ A full description of the statistical parameters including central tendency (e.g. means) or other basic estimates (e.g. regression coefficient) AND variation (e.g. standard deviation) or associated estimates of uncertainty (e.g. confidence intervals)
☐	☒ For null hypothesis testing, the test statistic (e.g. <i>F</i> , <i>t</i> , <i>r</i>) with confidence intervals, effect sizes, degrees of freedom and <i>P</i> value noted <i>Give P values as exact values whenever suitable.</i>
☒	☐ For Bayesian analysis, information on the choice of priors and Markov chain Monte Carlo settings
☒	☐ For hierarchical and complex designs, identification of the appropriate level for tests and full reporting of outcomes
☒	☐ Estimates of effect sizes (e.g. Cohen's <i>d</i> , Pearson's <i>r</i>), indicating how they were calculated

Our web collection on [statistics for biologists](#) contains articles on many of the points above.

Software and code

Policy information about [availability of computer code](#)

Data collection	Real time PCR data were collected with Bio-rad CFX v3.1 software.
Data analysis	Image J v1.53t; GraphPad Prism 9 (Dotmatics); Quantity One 1-D v4.6.8 (Bio-Rad)

For manuscripts utilizing custom algorithms or software that are central to the research but not yet described in published literature, software must be made available to editors and reviewers. We strongly encourage code deposition in a community repository (e.g. GitHub). See the Nature Portfolio [guidelines for submitting code & software](#) for further information.

Data

Policy information about [availability of data](#)

All manuscripts must include a [data availability statement](#). This statement should provide the following information, where applicable:

- Accession codes, unique identifiers, or web links for publicly available datasets
- A description of any restrictions on data availability
- For clinical datasets or third party data, please ensure that the statement adheres to our [policy](#)

Publicly available datasets used in this study includes PlantcircBase v7.0 (<http://ibi.zju.edu.cn/plantcircbase/index.php>); Bedtools v2.28.0 (<https://bedtools.readthedocs.io/en/latest/index.html>); The DRIP-seq data used for R-loop identification in this study had been submitted by Fang et al. (accession number GSE111944). Fang, Y. et al. Characterization of functional relationships of R-loops with gene transcription and epigenetic modifications in rice. Genome Res 29, 1287-1297 (2019). Pol II ChIP-seq data used in this study was available at xxx (ref??) RNAseq data sets of NIL-sdh and Nip have been stored in Sequence Read

Research involving human participants, their data, or biological material

Policy information about studies with [human participants or human data](#). See also policy information about [sex, gender \(identity/presentation\), and sexual orientation](#) and [race, ethnicity and racism](#).

Reporting on sex and gender	n/a
Reporting on race, ethnicity, or other socially relevant groupings	n/a
Population characteristics	n/a
Recruitment	n/a
Ethics oversight	n/a

Note that full information on the approval of the study protocol must also be provided in the manuscript.

Field-specific reporting

Please select the one below that is the best fit for your research. If you are not sure, read the appropriate sections before making your selection.

☒ Life sciences ☐ Behavioural & social sciences ☐ Ecological, evolutionary & environmental sciences

For a reference copy of the document with all sections, see nature.com/documents/nr-reporting-summary-flat.pdf

Life sciences study design

All studies must disclose on these points even when the disclosure is negative.

Sample size	No sample-size calculation was performed. Sample sizes were chosen as comparable to typical sizes of this type of study. Fine mapping was based on a set of >3,000 BC3F2 population bred from backcrossing NIL-sdh x indica variety 9311; We collected samples large enough in size >=100 seedlings per DNA pool in bulked segregant analysis ; and >=20 seedlings for per repoterline for phenotyping statistics. The sample size was determined sufficient by considering the extent and consistency of measurable distinctions among various plant lines, as well as significance and effect size.
Data exclusions	No data were excluded from the analysis in the article.
Replication	All experiments were repeated at least 3 times using independent biological replicates and at least 3 technical replicates per experiment. All replicates were indicating the same conclusions.
Randomization	Samples were randomly allocated into experimental groups for analysis.
Blinding	Blinding was not applicable to this study as considered not useful for the nature of experiments performed here. Instead, at least three biological replicates were performed in each experiment.

Reporting for specific materials, systems and methods

We require information from authors about some types of materials, experimental systems and methods used in many studies. Here, indicate whether each material, system or method listed is relevant to your study. If you are not sure if a list item applies to your research, read the appropriate section before selecting a response.

Materials & experimental systems

n/a	Involved in the study
☐	☒ Antibodies
☒	☐ Eukaryotic cell lines
☒	☐ Palaeontology and archaeology
☒	☐ Animals and other organisms
☒	☐ Clinical data
☒	☐ Dual use research of concern
☐	☒ Plants

Methods

n/a	Involved in the study
☒	☐ ChIP-seq
☒	☐ Flow cytometry
☒	☐ MRI-based neuroimaging

Antibodies

Antibodies used	S9.6 antibody (Abcam ab234957); Anti-H3K27ac antibody (Abcam ab4729); Anti-H3K27me3 antibody (Abcam ab6002); Goat anti-mouse-HRP secondary antibody (Abcam ab205719)
Validation	All antibodies used in this study are commercially available and the data sheet of the antibodies were provided by the companies. Primary antibodies were used in the previous research as cited in the manuscript. S9.6 antibody: https://www.abcam.cn/products/primary-antibodies/dnarna-hybrid-antibody-s96-ab234957.html?productWallTab=ShowAll Anti-H3K27ac antibody: https://www.abcam.cn/products/primary-antibodies/histone-h3-acetyl-k27-antibody-chip-grade-ab4729.html Anti-H3K27me3 antibody: https://www.abcam.cn/products/primary-antibodies/histone-h3-tri-methyl-k27-antibody-mabcam-6002-chip-grade-ab6002.html Goat anti-mouse-HRP secondary antibody: https://www.abcam.cn/products/secondary-antibodies/goat-mouse-igg-hl-hrp-ab205719.html

Dual use research of concern

Policy information about [dual use research of concern](#)

Hazards

Could the accidental, deliberate or reckless misuse of agents or technologies generated in the work, or the application of information presented in the manuscript, pose a threat to:

No	Yes
☒	☐ Public health
☒	☐ National security
☒	☐ Crops and/or livestock
☒	☐ Ecosystems
☒	☐ Any other significant area

Experiments of concern

Does the work involve any of these experiments of concern:

No	Yes
☒	☐ Demonstrate how to render a vaccine ineffective
☒	☐ Confer resistance to therapeutically useful antibiotics or antiviral agents
☒	☐ Enhance the virulence of a pathogen or render a nonpathogen virulent
☒	☐ Increase transmissibility of a pathogen
☒	☐ Alter the host range of a pathogen
☒	☐ Enable evasion of diagnostic/detection modalities
☒	☐ Enable the weaponization of a biological agent or toxin
☒	☐ Any other potentially harmful combination of experiments and agents

Plants

Seed stocks	Common rice varieties Nipponbare, 9311, Gui649 were used for transformation, various population creation and phenotyping. The plants were growing in glasshouses or in the experimental field in Nanning of Guangxi province or Linshui of Hainan province, China.
Novel plant genotypes	The near isogenic sdh line (NIL-sdh) was generated in the background of Japonica rice Nipponbare (Nip) through backcrossing sdh to Nip for seven times. A population derived from crossing NIL-sdh with the indica variety 9311 was used for map-based cloning analysis. We generated 8 independent lines of transgenic rice harboring miR156dSDH and miR156sdh driven by 2423 bp promoter of MIR156D in the Nip background, respectively. The different lines exhibited similar phenotype as described in the manuscript. The construct of the STTM with 88bp spacer linker design for miR156 was transformed into Nip and the transgenic rice with highest transgene expression rate of 8 independent lines were crossed with NIL-sdh to generate NIL-sdh STTM156. The transgenic rice of OE-miR156dcr1 was created by importing mutant miR156 sequence into the genomic region of MIR156D from 1st exon to 5th exon instead of the original miR156 driven by the Ubiquitin promoter. Based on the expression levels, three out of 11 independent transgenic lines were selected for the subsequent experiments indicated in the manuscript. Line #1 and #6 of OE-miR156dcr1 were crossed with NIL-sdh to yield OE-miR156dcr1-1 NIL-sdh and OE-miR156dcr1-6 NIL-sdh. Similarly, overexpression of truncated linear ciMIR156D without 3rd intron (OE-156dw/t 3rd intron) was generated. To explore whether the back-splicing events necessary for ciMIR156Ds production would induce nonsense-mediated decay (NMD) on the pri-miR156d, we generated the mutant of Os07g0495900 by CRISPR-Cas9. All the homogenous transgenic lines and crossed lines were obtained at T2 and F2 generation respectively.
Authentication	All the mutant and transgenic rice materials in this study were genotyped by PCR according to their specific frequencies along with the primer by using selection.